

MC-1: Design and Development of AMTTP — A Four-Layer Real-Time Blockchain Compliance System

Overview: The Anti-Money Laundering Transaction Trust Protocol (AMTTP) evaluates blockchain transactions **before settlement**, producing deterministic Approve / Review / Escrow / Block outcomes within Ethereum’s 12-second block window. In contrast to conventional blockchain compliance workflows that are largely post-transaction and forensic in nature, AMTTP is designed for execution-time risk control before final settlement. I am the **sole architect and developer** of all principal components: ML pipeline, Solidity contracts, the orchestrator and 9 specialist microservices, and both Flutter and Next.js frontends.

Public Testnet Demonstration at amttp.com

[CLICK HERE — 3-MINUTE VIDEO DEMONSTRATION]

Public demo: **amttp.com** — Sepolia-backed environment — 1,220 unique visitors
— 38,320 requests — 5 Apr – 5 May 2026

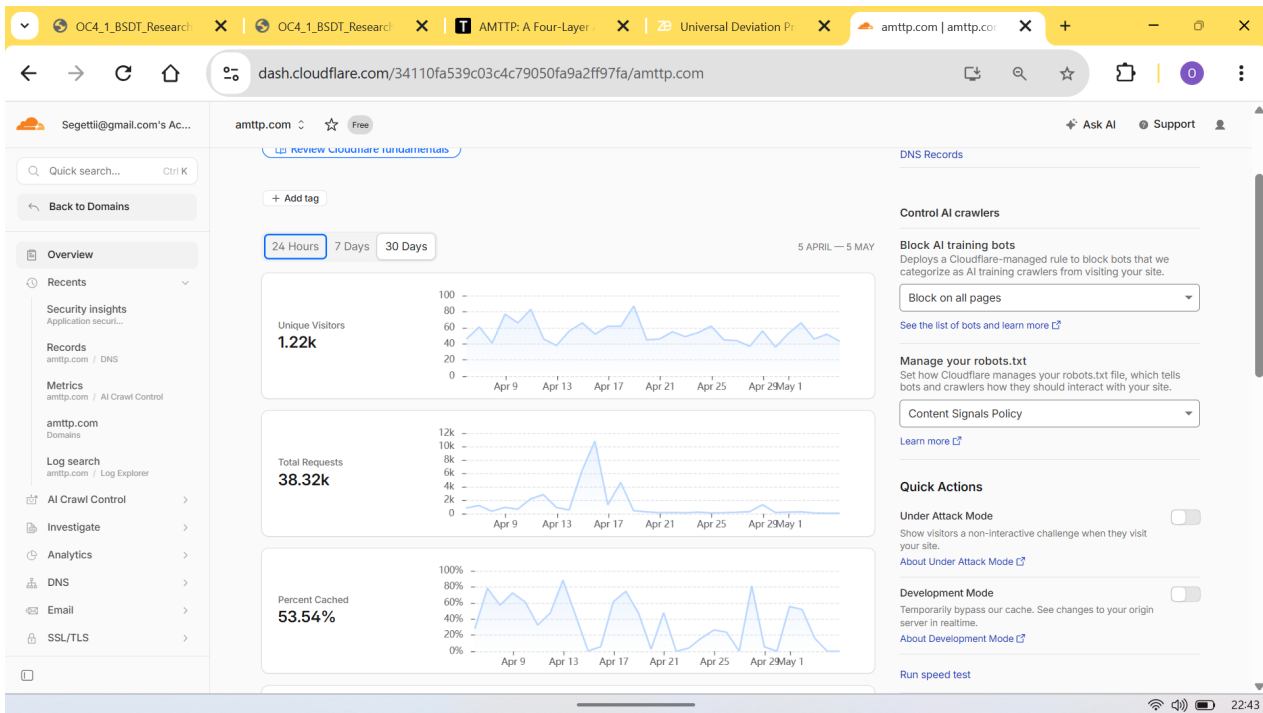


Figure 1: Cloudflare analytics — 1,220 unique visitors, 38,320 total requests, 53.54% cache-hit ratio over the 30-day window (5 April – 5 May 2026). Confirms sustained public traffic to the Sepolia-backed demonstration at amttp.com.

Key Technical Innovations

- **Pre-settlement enforcement:** compliance before EVM execution; unlike conventional forensic workflows, risk is evaluated before settlement.
- **Hybrid ML + graph scoring:** XGBoost fused with GraphSAGE GNN over live Memgraph entity graph.
- **Knowledge distillation:** CPU-portable student model; sub-12 s inference (Evidence 6).
- **ZK-SNARK privacy:** Groth16/BN254 proves KYC & sanctions on-chain without identity disclosure.
- **Deterministic decisions:** Approve/Review/Escrow/Block; full FCA/MiCA audit trail.
- **UUPS upgradeable protocol:** 5 contracts on OpenZeppelin proxies; zero-downtime upgrades.

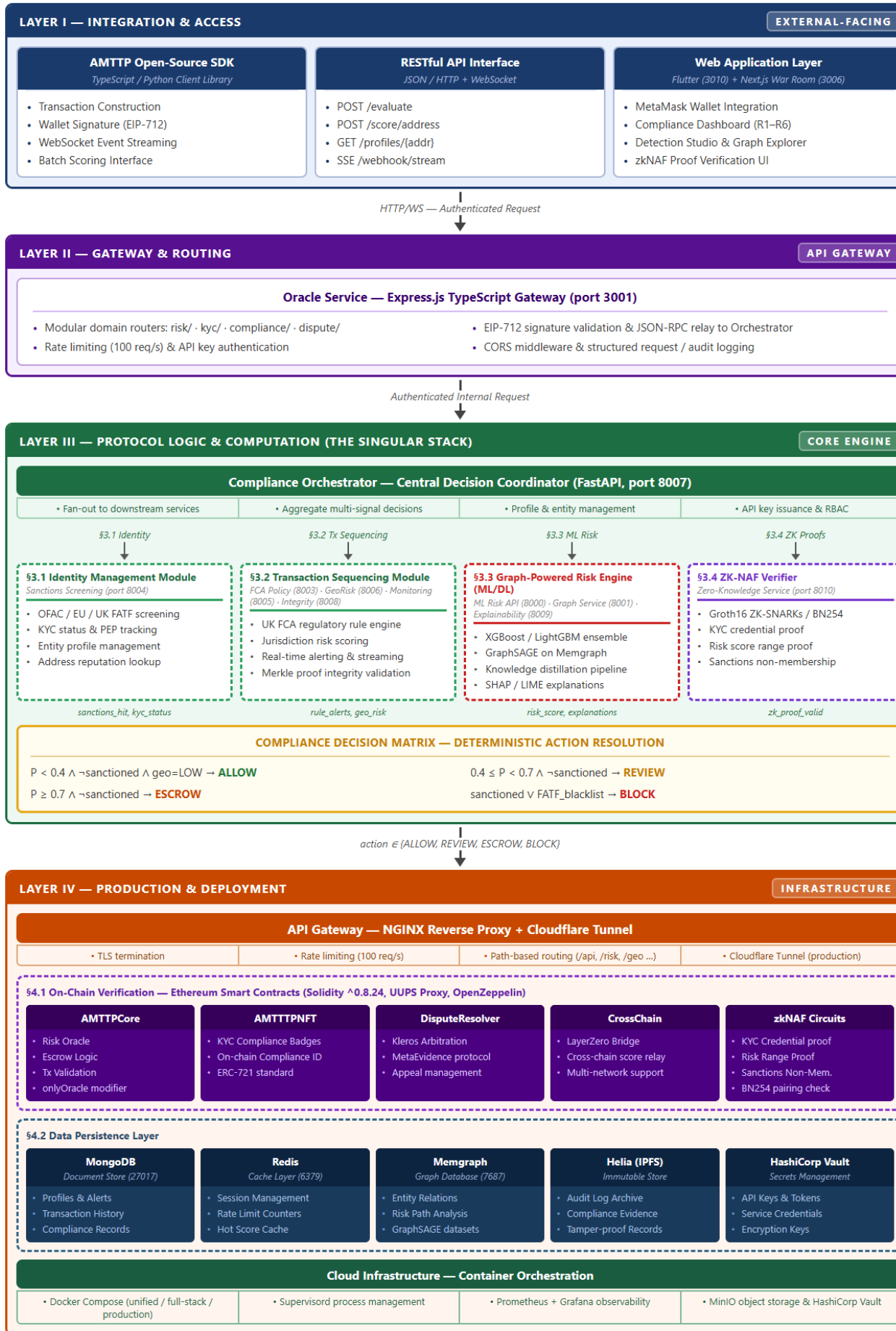


Figure 2: AMTTP four-layer architecture — Layer I: Frontends (Flutter :3010, Next.js :3006). Layer II: Oracle Gateway :3001. Layer III: FastAPI orchestrator + 9 specialist microservices + Deterministic Decision Matrix. Layer IV: 5 UUPS contracts, 17 Docker containers. Sole developer.

War Room Dashboard — Public Testnet at amttp.com

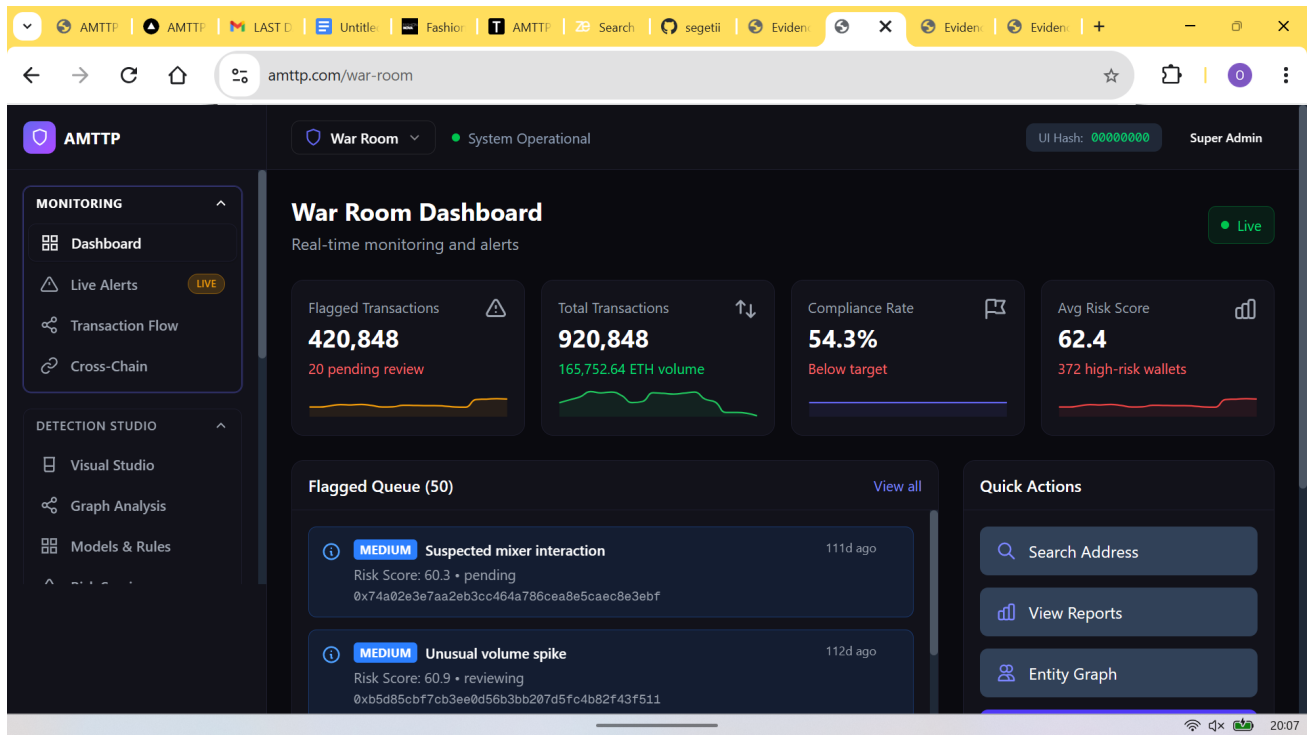


Figure 3: Public War Room dashboard in the AMTTP Sepolia testnet environment — 920,848 transactions processed, 420,848 flagged, compliance rate 54.3%, and average risk score 62.4. Wallet-format addresses in the flagged queue confirm sustained public operation rather than a static mock-up.

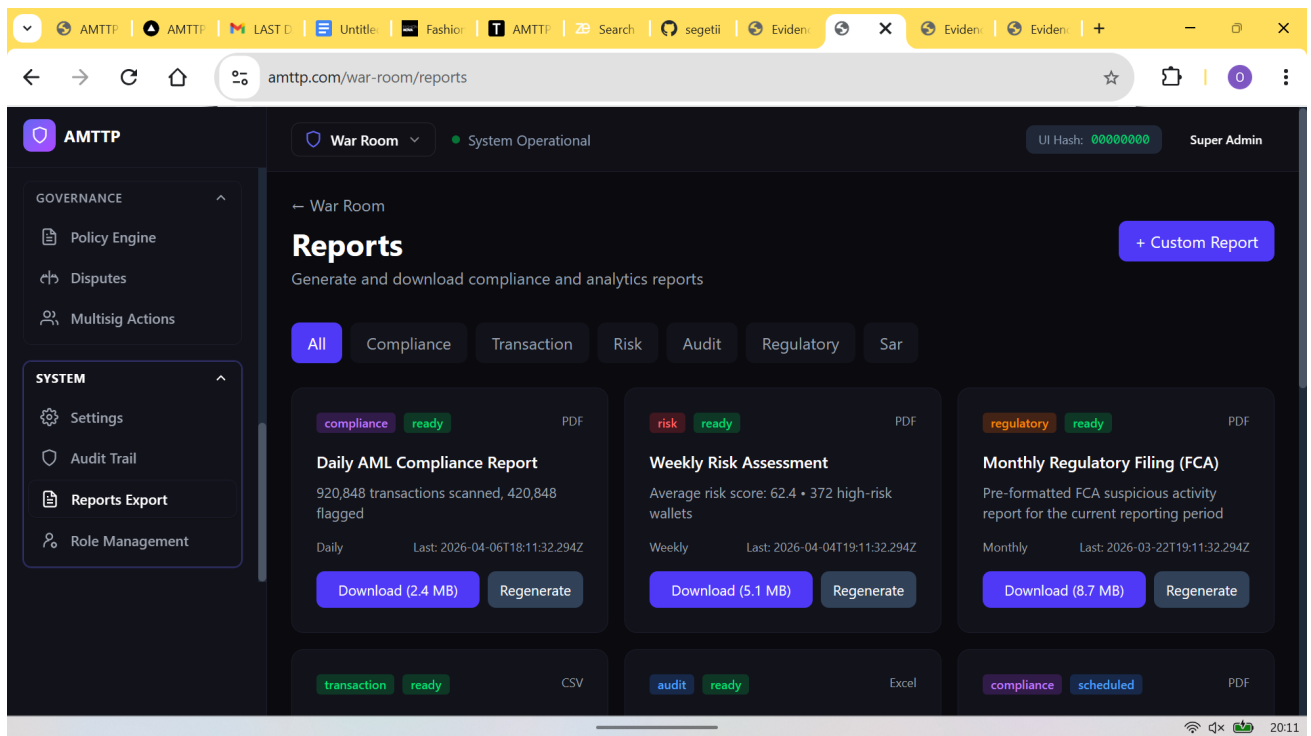


Figure 4: Reports Export — pre-generated Daily AML Compliance PDF (2.4 MB), Weekly Risk Assessment (5.1 MB), and Monthly FCA Suspicious Activity Report (8.7MB) demonstrating regulator-style reporting outputs generated by the system.